

Interatomic Potentials Benchmarking

This document details the methodology, execution, and analytical conclusions of our MACE-based neural network potential evaluation. The objective was to benchmark architectural complexity (spherical harmonics $L=0, 1, 2$) and thermodynamic data diversity (300K vs. 800K isolation) against a foundation baseline, measuring both static predictive accuracy and dynamic stability.

1. Methodology and Model Architectures

To rigorously evaluate the generalist-specialist trade-off in computational physics, we developed three distinct training paradigms using 1,200 fully equilibrated AIMD frames (NVT/NPT ensembles, 300K–800K):

- **Universal Specialists (From-Scratch):** Trained exclusively on our target dataset across three equivariant architectures: $L=0$ (Scalar), $L=1$ (Vector), and $L=2$ (Full Tensor).
- **The Adapted Expert (Finetuned L_0):** A baseline Small foundation model finetuned using a multi-head approach, integrating 1,000 background frames from the Materials Project to prevent catastrophic forgetting while adapting to our specific thermodynamic states.
- **Temperature-Isolated Models (Data Diversity):** Two $L=0$ models trained exclusively on isolated thermal boundaries—one strictly at 300K (Cold Model) and one strictly at 800K (Hot Model)—to test thermal extrapolation limits.

2. Static Predictive Accuracy (RMSE & Parity)

Static inference on the validation set quantified the direct impact of higher-order tensor mathematics on force and energy predictions.

- **Architectural Scaling:** Increasing mathematical complexity yielded strict monotonic improvements in predictive accuracy. The Universal L_2 model achieved the lowest errors (22.0 meV/Å for forces, 1.2 meV/atom for energy), confirming that full tensor spherical harmonics capture complex atomic environments significantly better than L_0 scalar math (34.0 meV/Å).
- **Foundation vs. Specialist:** The Finetuned L_0 model exhibited a higher static force error (59.2 meV/Å) compared to the from-scratch Universal L_0 . This quantifies the exact cost of the generalist-specialist trade-off: the model sacrifices a degree of local precision to retain broad chemical knowledge.
- **Thermal Bias:** The isolated 300K model exhibited massive static errors (120.0 meV/Å) when evaluated across the diverse validation set, signaling deep structural overfitting to low-energy geometries.

3. Dynamic Stability: 600K Langevin Molecular Dynamics

To evaluate physical viability, all models were subjected to 5,000 steps of Langevin molecular dynamics at a target temperature of 600K using the ASE environment. Stability was assessed via Potential Energy drift and the Radial Distribution Function, $g(r)$.

- **Extrapolation Failure:** The Cold Model (300K) suffered catastrophic physical breakdown. Potential energy fluctuations exploded after 2,000 steps, and its $g(r)$ plot demonstrated severe unphysical nuclear overlap near 0.0 Å, missing the primary 1.1 Å coordination peak entirely. This proves ML potentials cannot reliably extrapolate thermal kinetics upward.
- **Interpolation Resilience:** The Hot Model (800K) survived the 600K simulation, accurately tracking the primary coordination peaks. This indicates that interpolating down to lower thermal-kinetic states is computationally safer than upward extrapolation.
- **Structural Integrity of Foundation Models:** Despite its higher static RMSE, the Finetuned L0 model maintained perfectly stable potential energy dynamically. Its $g(r)$ peaks overlapped seamlessly with the Universal models, proving that foundation models can accurately hold crystal structures at high temperatures without requiring computationally heavy tensor operations.

4. Strategic Conclusions

1. **Tensor Math for Precision:** Full tensor mathematics ($L=2$) are required for absolute static precision, but scalar ($L=0$) architectures are sufficient to maintain structural integrity in dynamic simulations if the thermodynamic training data is diverse.
2. **Asymmetric Thermal Vulnerability:** Neural network potentials exhibit severe asymmetric vulnerability to temperature; they can safely interpolate to cooler states but undergo catastrophic geometric failure when extrapolating to higher thermal energies.
3. **Foundation Viability:** Multi-head fine-tuning of foundation models successfully preserves global chemistry knowledge without compromising the localized thermodynamic stability of the target material, providing a highly efficient middle ground for molecular dynamics.